



Complex Numbers · Mark Scheme

Core Pure 1 Chapter 1: Complex Numbers. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Arithmetic with complex numbers

- 1 (a) $i^{11} = (i^2)^5 i = -i$ B1; (b) $\sqrt{-49} = 7i$ B1.
- 2 $3 + 12i - 2i - 8i^2$ M1; $= 11 + 10i$ A1.
- 3 $z^2 = 4 - 4i + i^2$ M1; $= 3 - 4i$ A1; $z^3 = (3 - 4i)(2 - i) = 2 - 11i$ A1.
- 4 $a^2 - b^2 = 5$ and $2ab = -12$ M1; $b = -\frac{6}{a}$, so $a^4 - 5a^2 - 36 = 0$ A1; $(a^2 - 9)(a^2 + 4) = 0$, $a^2 = 9$ M1; $a = 3, b = -2$ or $a = -3, b = 2$ A1.
- 5(a) $w^2 = 1 + 4i + 4i^2$ M1; $= -3 + 4i$ A1.
- 5(b) $(-3 + p + q) + (4 + 2p)i = 0$ M1; imaginary part: $p = -2$ A1; real part: $q = 5$ A1.
- 5(c) $w^4 = (-3 + 4i)^2 = -7 - 24i$ B1; $w^4 + kw^2 = (-7 - 3k) + (-24 + 4k)i$, so $-24 + 4k = 0$ M1; $k = 6$ A1.

Section B – Conjugates and division

- 6 $z^* = 5 + 3i$ B1; $zz^* = 25 + 9 = 34$ B1.
- 7 $\frac{(3 + 4i)(1 + 2i)}{(1 - 2i)(1 + 2i)} = \frac{-5 + 10i}{5}$ M1; $= -1 + 2i$ A1.
- 8 $z = \frac{3 + 5i}{1 + i}$ M1; $= \frac{(3 + 5i)(1 - i)}{2} = \frac{8 + 2i}{2}$ M1; $z = 4 + i$ A1.
- 9 Let $z = x + yi$, $z^* = x - yi$ M1; $x + yi + 2xi + 2y = (x + 2y) + (2x + y)i$ A1; $x + 2y = 7$, $2x + y = 8$ M1; $z = 3 + 2i$ A1.
- 10(a) $2 - 2i + 3i - 3i^2$ M1; $= 5 + i$ A1.
- 10(b) Multiply numerator and denominator by $1 + i$ M1; $\frac{-1 + 5i}{2}$ A1; $= -\frac{1}{2} + \frac{5}{2}i$ A1.
- 10(c) $\frac{(2 + \lambda + 3i)(1 + i)}{2} = \frac{(\lambda - 1) + (\lambda + 5)i}{2}$ M1; real part = 0: $\lambda - 1 = 0$ M1; $\lambda = 1$ A1.

Section C – Quadratics with complex roots

- 11 $z^2 = -16$ M1; $z = \pm 4i$ A1.
- 12 $(z + 3)^2 = -4$ or formula M1; $z = -3 \pm 2i$ A1.
- 13 $z = \frac{2 \pm \sqrt{4 - 40}}{4}$ M1; $= \frac{2 \pm 6i}{4}$ A1; $z = \frac{1}{2} \pm \frac{3}{2}i$ A1.
- 14 Other root is $3 + 4i$ (real coefficients) B1; sum of roots $6 = -b$ M1; $b = -6$ A1; product $(3 - 4i)(3 + 4i) = 25 = c$ A1.
- 15(a) $b^2 - 4ac = 16 - 4k < 0$ M1; $k > 4$ A1.
- 15(b) $z = \frac{4 \pm \sqrt{16 - 52}}{2}$ M1; $= \frac{4 \pm 6i}{2}$ A1; $z = 2 \pm 3i$ A1.
- 15(c) $(2 + 3i)^2 + (2 - 3i)^2 = (-5 + 12i) + (-5 - 12i)$ M1; $= -10$ A1.

Section D – Cubics and quartics

- 16 $2 - i$ B1; $(z - 2 - i)(z - 2 + i) = (z - 2)^2 + 1 = z^2 - 4z + 5$ B1.
- 17 $1 - 1 + 4 - 4 = 0$ B1; $(z - 1)(z^2 + 4) = 0$ M1; $z = 1, \pm 2i$ A1.



18 $1 - 2i$ is a root *B1*; quadratic factor $z^2 - 2z + 5$, and $z^3 - 5z^2 + 11z - 15 = (z^2 - 2z + 5)(z - 3)$ *M1*; roots $1 - 2i$ and 3 *A1*.

19 $-3i$ is also a root, so $z^2 + 9$ is a factor *B1*; $(z^2 + 9)(z^2 + az + b)$: compare coefficients *M1*; $z^2 - 2z + 5$ *A1*; $z = \pm 3i$, $1 \pm 2i$ *A1*.

20(a) $2 + 3i$ *B1*.

20(b) $(z - (2 - 3i))(z - (2 + 3i))$ *M1*; $= (z - 2)^2 + 9$ *M1*; $= z^2 - 4z + 13$, so it is a factor *A1*.

20(c) $f(z) = (z^2 - 4z + 13)(z^2 + az + b)$, compare coefficients or divide *M1*; other factor $z^2 + 4$ *A1*; $z^2 = -4$ *M1*; $z = 2 \pm 3i$, $\pm 2i$ *A1*.

Roar-level challenge

21 $(a + bi)^2 = -5 + 12i$: $a^2 - b^2 = -5$, $2ab = 12$ *M1*; square roots $\pm(2 + 3i)$ *A1*; discriminant $(4 - i)^2 - 4(5 - 5i) = -5 + 12i$, so $z = \frac{(4 - i) \pm (2 + 3i)}{2}$ *M1*; $z = 3 + i$ *A1*; $z = 1 - 2i$ *A1*.